

Total No. of Questions : 6]

SEAT No. :

P478

[Total No. of Pages : 2

APR - 18/BE/Insem. - 21

B.E. (Mechanical)

INDUSTRIAL ENGINEERING

(2012 Course) (Elective - III) (Semester - II) (402049C)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) *Answer Q1 or Q2, Q3 or Q4, Q5 or Q6.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Figures to the right side indicate full marks.*
- 4) *Assume suitable data, if necessary.*
- 5) *Answers should be written in one answer book.*

Q1) a) Discuss the Scope & objectives of Industrial Engineering. **[6]**

b) Write a short note on System approach. **[4]**

OR

Q2) a) Explain briefly the Productivity improving methods for organization. **[4]**

b) The following information available for a company given below : **[6]**

- i) Daily working hrs = 8
- ii) No. of working days in week = 6
- iii) No. of operators = 20
- iv) Std. hours per units production = 4

During a particular week

- v) No. of units produced = 48
- vi) Absentee man days = 40
- vii) Idle time due to load shedding = 30 May days.

Find : Absenteeism % productivity, Labour utilization percentage, Productive efficiency of labour, and overall productivity of labour in terms of units produced/week/employee.

P.T.O.

Q3) a) What are the various allowances used. Why allowances are considered in time study? [6]

b) Define work sampling. Explain in detail work sampling Procedure. [4]

OR

Q4) a) What are therbligs? Give any five therbligs with symbols. [6]

b) Difference between work measurement & work study. [4]

Q5) a) Write short notes on : (Any two) [4]

i) PMTS.

ii) Multiple Activity chart.

iii) MOST.

b) The work study engineer carried out the work sampling study. The following observation were for a machine shop. [6]

i) The time duration of study = 120 hrs.

ii) Total no. observation = 7000

iii) No. working activities = 1200

iv) Ratio between manual to a machine elements 2 : 1

v) Average rating factor = 120%

vi) Total no. of jobs produced during study = 800 units

vii) Rest and personal time for the job.

OR

Q6) a) What is time study? State their limitations. [4]

b) The following data refers to a sampling study of production of one component. [6]

i) Duration of data collection = 5 days, 8 hrs per day.

ii) No. of Operators = 10.

iii) Allowances given for the process = 5%.

iv) Production quantity in 5 days = 6000 components.

v) Sampling data collected.

Days	1	2	3	4	5
No. of Observations	230	240	200	180	225
Occurance of activity	200	190	170	150	210

